# Determining the Specific Heat of Gadolinium

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Abstract goes here.

## I. INTRODUCTION

The heat capacity of gadolinium (Gd), the $64^{\text{th}}$ element is of interest due to its ferromagnetic properties. Gadolinium is known to have a spontaneous magnetic moment below its Curie temperature $T_C$, which means that there exist macroscopic regions in which the atomic magnetic moments are aligned. As we will soon see, this spontaneous magnetic moment has interesting thermodynamic implications.

### A. Ising Model Thermodynamics

In the Ising thermodynamic model, a given material can be modeled as a grid or lattice with each vertex representing the location of an atom with spin $\pm 1/2$. Each vertex thus has a potential energy if it is antiparallel with its neighboring spins[1], and negative when it is parallel. Assuming our material is not in the presence of an external magnetic field, if it goes from having no net magnetization to ferromagnetic, we would expect to observe a spontaneous drop in system energy. It is this instantaneous drop in energy that results in a sharp kink in the curve of energy versus temperature.

The heat capacity of a system describes its ability to increase in energy without also experiencing a large change in temperature. This value is also often reported to be per unit mass. From this we conclude that excluding any external sources or drains of energy, the heat capacity can be represented as the derivative of a energy temperature plot. The kink at $T_C$ in the aforementioned plot of energy vs. temperature results in a discontinuity of the heat capacity versus temperature. Given that we gain a discontinuity in the derivative of our order parameter, the behavior at and around $T_C$ represents a second order transition[1].

## II. APPARATUS AND PROCEDURE

### A. Gadolinium Apparatus

Our gadolinium sample was a cylinder $11 \pm 2mm$ tall with a $7 \pm 2mm$ radius, and was placed within a vacuum chamber. Attached to the gadolinium sample and also within the vacuum chamber was a heating coil, which we could use to increase the sample energy. In contact with the gadolinium was a brass post smaller than the sample, and the post itself was attached to a large aluminum cylinder which was outside of the vacuum region.

To measure the temperature of our sample, we used Vishay thermistors and a spline to interpolate the manufacturer's temperature vs. resistance data points. To measure the heat capacitance at various temperatures, we first had to bring the sample to a steady temperature, which we did by bringing it and the aluminum stage into equilibrium. The aluminum stage could be cooled or heated using a thermoelectric cooler attached to its base. Its cooling performance could be further improved by cryogenically cooling its base.

We utilized two Keithley digital multimeters via a single serial GPIB line to take time series measurements that allowed us to better understand the temporal behavior of the system. The Keithley 2000 multimeters were limited in their measurement frequency, allowing us to collect resistance measurements for both the sample and the stage every $\approx 1.5s$. However, this was not a limiting factor in our experiment as any meaningful changes in temperature happened over the course of minutes.

### B. Thermoelectric Cooler "P"ID Controller

Our heat capacity measurements involved bringing the sample to a steady temperature, heating it with the coil, and calculating the amount of energy that went from the coil to the sample. Given that both the sample and coil are in vacuum, the energy that goes into the sample is simply the heater power multiplied by the on time. However, the gadolinium dissipates some of its heat through the brass rod to the aluminum stage. We mitigate this by starting them at the same temperature.

Keeping the aluminum stage at a fixed temperature required writing a control loop. Given that we could not switch the polarity of the thermoelectric cooler, we had to choose between active cooling or heating for a given trials. For target temperatures above room temperature we used active heating and simply allowed it passively cool if we overshot the target. The same method worked vice-versa for target temperatures below room temperature. The downside of this is that we get asymmetric performance when balancing about our target temperature, resulting in oscillations that are challenging to dampen.

As mentioned previously, meaningful changes in temperature occurred on the order of minutes, which has both benefits and drawbacks. The system had plenty of time to react to errors, but also responded slowly. The slow response additionally made it challenging to evalu-

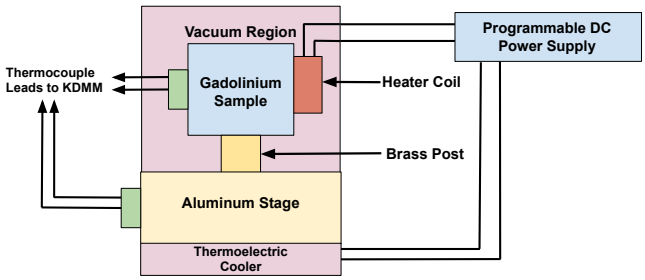



FIG. 1. The apparatus of our g gadolinium sample and associated equipment.

ate the performance of a given controller parameter. We began with a proportional controller tuned to the best of our abilities, and added a quadratic term to reduce response at small errors. Ultimately, many oscillations remained, though they were often centered near the target temperature, with amplitudes no greater than $0.2°C$

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[1] Richard Fitzpatrick, The Ising Model, University of Texas at Austin, 2006 [**?** ] Grzegorz Bednarz, Heat Capacity of Gadolinium near the Curie temperature, Physical Review B, 1993. [**?** ] P. C. Lancheste, The Critical Specific Heat of Gadolinium, Journal of Magnetism and Magnetic Materials, 1980.